

Rajat Dhanraj Savdekar

Los Angeles, CA | (213) 574-8894 | rsavdeka@usc.edu | LinkedIn | github.com/RDX-Rajat-Savdekar | rajatsavdekar.dev

EDUCATION

University of Southern California

Masters of Science in Computer Science

Relevant Coursework: Algorithms, Web Technology, Data Science/ML, Databases, Multimedia System

Los Angeles, California

Aug 2024 – **May 2026**

GPA: 3.6

University of Mumbai

Bachelor of Engineering in Computer Engineering

Relevant Coursework: Computer Networks, Software Engineering, Data Structures, AR/VR

Mumbai, India

Aug 2019 – May 2023

GPA: 3.7

SKILLS

• **Languages:** Python, Java, JavaScript, C++, C#
• **Frontend:** React, Angular, HTML, CSS, Vite, Next.js
• **ML / AI Tools:** PyTorch, OpenAI CLIP, Embeddings, Similarity Metrics, Multimodal Learning
• **Performance & UX:** Animation / Visualization (Manim), Responsive Design, Cross-browser Compatibility

• **Databases:** MySQL, PostgreSQL, MongoDB, SQL, Vector
• **Backend:** Node.js, Flask, OpenAPI, REST APIs, PDF.js
• **Cloud / DevOps:** AWS, Azure, Google Cloud Platform, Firebase, Vercel, Docker, GitHub Actions, CI/CD
• **Tools & Workflow:** Git, Figma, Jest, Postman, Swagger

WORK EXPERIENCE

Teaching Assistant

USC Summer Programs

May 2025 – July 2025

Los Angeles, California

- Led Arduino-based coding sessions for ~160 students, improving programming logic and circuit design.
- Co-designed energy and electronics projects for over 100 participants, integrating coding with sustainability concepts.
- Mentored student teams in engineering challenges, fostering teamwork, iterative problem-solving, and confidence.
- Coordinated with 54 SoCalGas and USC staff volunteers to deliver seamless hands-on STEM workshops.

Software Engineer

Jalgaon Fruits Sales Cooperative

May 2023 – May 2024

Jalgaon, India

- Digitized procurement and inventory workflows, replacing manual logs with a **web-based system** for 150 daily users, reducing errors and enabling faster farmer payments.
- Refactored legacy PHP modules into **Flask microservices**, improving responsiveness, simplifying maintenance, and strengthening reliability of core workflows.
- Introduced **Docker containerization** and **CI/CD pipelines (GitHub Actions)** to stabilize deployments and accelerate release cycles by ~30%, ensuring reliability during peak harvest periods.
- Built **React + Flask dashboards** with real-time analytics, providing **data-driven insights** that reduced spoilage and improved transparency for 150+ daily users.

PROJECTS

Resume Tailor (Free) – Next.js, React, Tailwind CSS, PDF.js, Vercel

[GitHub]

- Built an open-source tool that tailors resumes to job descriptions using keyword extraction and document comparison.
- Implemented PDF parsing with pdf.js and designed a responsive UI for side-by-side resume vs JD review.
- Deployed on Vercel with a modular React + Tailwind architecture for maintainability and fast iteration.

Retrieval-Augmented Generation System – Python, LLMs, Vector Search

[GitHub]

- Engineered a modular **RAG pipeline** integrating vector retrieval with LLMs for context-aware generation.
- Improved query precision through similarity search with relevance ranking, enhancing downstream response quality.

Multi-modal Search Engine Using CLIP – Python, PyTorch, OpenAI CLIP

[GitHub]

- Implemented a search engine leveraging joint **text and image embeddings** for natural language image retrieval.
- Evaluated system with precision@5 and recall metrics, refining cosine similarity thresholds for balanced relevance and diversity.

Manim-DSA-SD-Concepts – Python, Manim

[GitHub]

- Designed and developed animated visualizations of core **data structures, algorithms, and system design concepts**, aiding conceptual clarity for learners.
- Engineered a modular rendering pipeline with caching and lazy loading, reducing build time by ~30%.

Typing Speed Web App – React, JavaScript

[GitHub]

- Built a responsive typing-test web app delivering real-time speed/accuracy feedback with adjustable difficulty modes.
- Optimized UI for cross-browser compatibility and mobile responsiveness, improving accessibility for 50+ test users.

Image Texture Analysis with GLCM – MATLAB, Image Processing

[Publication]

- Developed a GUI tool to compute texture features (contrast, entropy, homogeneity, correlation) from images.
- Cited in 10+ IEEE and Springer Nature papers, demonstrating research impact and academic relevance.